CUSTOMER RETENTION ANALYTICS

Customer ChurnSense

AI System Using Machine Learning

DATA

MODEL

DECISION

Historical customer information as an early churn signal

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Earlier churn signals can create time for customer review

DECISION WINDOW

01 Customer history
Customer profile, relationship, billing and service records

02 Risk prediction
The classifier assigns a likely stay or churn label

03 Customer review
A business team can examine the account context

04 Illustrative action
Engagement, service recovery or a suitable offer

AVAILABLE INFORMATION

PROFILE
Gender and age

RELATIONSHIP
Tenure

BILLING
Monthly and total charges

CONTRACT & SERVICE
Contract type, internet service and related attributes

BINARY TARGET

0 Likely to stay

1 Likely to churn

~1,000
static records described in the synopsis

Boundary
The output supports review. It does not explain why a customer leaves.

Source: Master synopsis, sections A-D, H-I

The documented workflow separates model training from future prediction

TRAINING PATH

1 Inspect	2 Prepare	3 Encode	4 Split	5 Scale	6 Train	7 Evaluate
shape, info, describe	fill missing values	convert categories	80% train, 20% test	fit on training data	Logistic Regression	metrics on test data
data.info()	*fillna()*	*LabelEncoder()*	*train_test_split()*	*StandardScaler()*	*model.fit()*	*classification metrics*

REUSE PATH

Persisted artefacts	New customer record	Transform	Predict
Saved model and scaler, as documented	Same fields, order and category treatment	Reuse the saved preprocessing logic	Return likely stay or churn

CONTROL POINTS

Fit scaling on training data
Reuse transformations for test and new records

Keep test data independent
Prevent evaluation data from influencing learning

Preserve the input schema
Maintain feature order and categorical mappings

Source: Master synopsis, sections D-G and J

Logistic Regression produces the class, while error analysis explains its meaning

MODEL LOGIC

Customer features

Tenure, charges, contract and service attributes

Weighted combination

$p = 1 / (1 + e^{-z})$

WHY THIS BASELINE

It fits a binary classification task and remains comparatively interpretable.

ERROR MATRIX

Actual / predicted	STAY	CHURN
ACTUAL STAY	**Correct stay** No alert needed	**False alert** Unnecessary review
ACTUAL CHURN	**Missed churner** Risk remains unseen	**Correct alert** Review can begin

METRIC INTERPRETATION

Accuracy	Share of all predictions that are correct
Precision	Reliability of churn alerts
Recall	Coverage of actual churners
F1	Balance of precision and recall
Support	Records evaluated in each class

The master synopsis does not report numerical model results.

Classification shows association, not the cause of churn.

Source: Master synopsis, sections E-G; Scikit-learn documentation for metric concepts

Operational use requires stronger validation and a process around each prediction

ILLUSTRATIVE BUSINESS WORKFLOW

1 **Prediction**
Likely stay or churn

2 **Account review**
Check customer context

3 **Response**
Engagement, service recovery or offer

4 **Outcome capture**
Record what happened

CAPABILITY GAP

DIMENSION	DOCUMENTED STATE	WHY IT MATTERS	IMPROVEMENT PATH
Data	Small, static dataset; limited attributes	May not generalise to a different customer base	Refresh data and add complaints or usage patterns
Model	Logistic Regression; Label Encoding	One baseline cannot establish the best approach	Compare other classifiers; use One-Hot Encoding where appropriate
Validation	80:20 hold-out evaluation	One split provides limited evidence of stability	Add cross-validation and hyperparameter tuning
Delivery	Offline data; no live CRM connection	Predictions remain separate from daily operations	Package an application, then integrate refreshed data

Conclusion: the synopsis documents a reusable churn prediction approach; dependable business impact remains to be established.

Source: Master synopsis, sections H-J

Technical environment and methodology reference

TECH STACK

Python 3.10	Programming language
NumPy + Pandas	Data preparation
Matplotlib + Seaborn	Exploratory analysis
Scikit-learn	Preprocessing, modelling and evaluation
Pickle	Documented artefact persistence

ENVIRONMENT

Windows 11; Google Colab / VS Code / Render

Intel i3 or higher; 4 GB RAM or higher; 20 GB free disk space

TESTING AND LIMITS

DOCUMENTED TESTING

80:20 hold-out evaluation
Accuracy, confusion matrix and classification report
Manual prediction with sample customer records

PROPOSED ROBUSTNESS CHECKS

Missing inputs and unseen categories
Feature order and schema consistency
Persistence and reload consistency

METHODOLOGICAL BOUNDARIES

The synopsis does not specify a churn horizon or exact Customer ID treatment. Predictions indicate association, not causation or certainty.

SOURCES Master synopsis; Scikit-learn, Pandas, NumPy, Matplotlib and Seaborn documentation; Kaggle reference; Sebastian Raschka, Python Machine Learning

Appendix source: Master synopsis, sections F-G and K

Thank you